# Phase–Flux Field (PFF): Axiomatic Substrate for Wave Confinement Theory

Zero–Wave Invariance, Finite–$k$ Lyapunov Band–Pass, Shell Quantization, and $D_4 \to$ Continuum

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**Abstract**

The *Phase–Flux Field* (PFF): a wave–first substrate specified only by observables—energy density $u(\mathbf{x}, t)$, energy flux $\mathbf{S}(\mathbf{x}, t)$, and phase $\theta(\mathbf{x}, t)$. Two axioms, local conservation and null (lightlike) flow, define the kinematics without EM, particle, or gauge semantics. The uniform *Zero Wave* is reserved for the seed state $\mathrm{ZW}_0$. A Lyapunov band–pass rail with linear growth $\sigma(k) = r + ak^2 - bk^4$ selects a finite ring and saturates via the $-\beta|A|^2 A$ term. With saturating nonlinearity gives IR/UV stability, finite–band growth, and locked spirals that wind into $2\pi r$ shells. We provide a discrete 4–node primer (with $D_4$ covariance, winding, and stress from folds), its continuum limit, a shell–quantization rule, and the zero–wave transform $T_0$ that connects PFF to standard thermodynamic reductions (Swift–Hohenberg / complex Ginzburg–Landau). This note is a conceptual precursor to WCT.

## 1 Scope and Philosophy (Pre–WCT)

WCT builds matter–like structures from *folding waves with Lyapunov descent.* Here we establish: (i) the substrate (PFF), (ii) the seed state (Zero Wave $\mathrm{ZW}_0$), (iii) a minimal band–pass + saturation rail with a Lyapunov functional, and (iv) the geometry/topology for shells and windings. No EM vocabulary is required; everything is in terms of observables.


Figure 1: Zero Wave ($\mathrm{ZW}_0$): uniform seed state.

## 2 PFF: Observables and Axioms

**Domain and observables.** Let $\Omega \subset \mathbb{R}^d$ be the spatial domain. The observable fields are

$$u(\mathbf{x},t) \geq 0, \qquad \mathbf{S}(\mathbf{x},t) \in \mathbb{R}^d, \qquad \theta(\mathbf{x},t) \in (-\pi,\pi].$$

**Axiom 1** (Local conservation)**.**

$$\partial_t u + \nabla\cdot\mathbf{S} = 0. \tag{1}$$

**Axiom 2** (Causality (null bound))**.** *There exists a fixed causal speed $c > 0$ such that*

$$|\mathbf{S}| \leq c\,u. \tag{2}$$

*When $u > 0$ and $|\mathbf{S}| = c\,u$ (null case), define the local energy–flow direction $\hat{\mathbf{n}} := \mathbf{S}/|\mathbf{S}|$.*

**Definition 2.1** (Phase–flux constitutive law)**.** The phase $\theta$ fixes the flux by

$$\mathbf{S} := u\,\nabla\theta, \tag{3}$$

so that (1) becomes $\partial_t u + \nabla\cdot(u\nabla\theta) = 0$. In nondimensional units the bound (2) is $|\nabla\theta| \leq c$ (with equality in the null case).

**Definition 2.2** (Zero Wave (seed state))**.** The *Zero Wave* $\mathrm{ZW}_0$ is the uniform state $u = u_0 > 0$ and $\nabla\theta = 0$ (up to tiny IR–damped fluctuations). The term "Zero Wave" is used only for this seed.

**Definition 2.3** (Winding and shells)**.** For any closed loop $\gamma$, the phase winding number is

$$m = \frac{1}{2\pi}\oint_\gamma \nabla\theta \cdot \mathrm{d}\boldsymbol{\ell} \in \mathbb{Z}.$$

Let $k := |\nabla\theta|$ and, in polar coordinates, $k_r(r) := \hat{\mathbf{r}}\cdot\nabla\theta$. Radial shells close when

$$\oint k_r(r)\,\mathrm{d}r = 2\pi n, \qquad n \in \mathbb{Z}.$$

## 3 Discrete Primer: 4–Node Square, $D_4$ Symmetry, Stress from Folds

Label the square nodes $i \in \{1, 2, 3, 4\}$ (spacing $\ell$); at node $i$ store $(u_i, \theta_i)$. On edge $i \leftrightarrow j$ define direction $\hat{e}_{ij}$, edge energy $u_{ij} = \frac{1}{2}(u_i + u_j)$, and null flux

$$\mathbf{S}_{ij} = c\, u_{ij}\, \sin(\theta_j - \theta_i)\, \hat{e}_{ij}, \qquad |\mathbf{S}_{ij}| \le c\, u_{ij}. \tag{4}$$

**Node conservation:**

$$\dot{u}_i = -\frac{1}{\ell} \sum_{j \sim i} \mathbf{S}_{ij} \cdot \hat{e}_{ij}, \qquad \Delta t \le \ell / c \quad \text{(causal step)}. \tag{5}$$

**Plaquette winding (notation is explicit):**

$$m = \mathrm{round}\left( \frac{\mathrm{wrap}(\theta_2 - \theta_1) + \mathrm{wrap}(\theta_3 - \theta_2) + \mathrm{wrap}(\theta_4 - \theta_3) + \mathrm{wrap}(\theta_1 - \theta_4)}{2\pi} \right). \tag{6}$$

Here $\mathrm{wrap}(\phi)$ denotes the unique representative of $\phi$ in $(-\pi, \pi]$; $\mathrm{round}(\cdot)$ denotes nearest integer. If all $u_i > 0$, $m$ is conserved; it can change only if some amplitude vanishes along a bridge (unwind event). All rotations/reflections of the square ($D_4$) preserve the form of Eqs. (4) to (6).

**Discrete stress from folds.** On the square cell,

$$\mathbf{T}_{\text{cell}} = \sum_{(i,j)\ \text{edges}} u_{ij}\, \hat{e}_{ij} \otimes \hat{e}_{ij}, \tag{7}$$

so phase bends (changes in $\theta$) rotate energy–carrying edge directions; $\nabla \cdot \mathbf{T}_{\text{cell}}$ gives net momentum flux into the cell.

## 4 Continuum Rail and Lyapunov Descent

Let $A(\mathbf{x}, t) = \sqrt{u}\, e^{i\theta}$ be the complex envelope. The PFF rail is

$$\partial_t A = \underbrace{(r - a\,\Delta - b\,\Delta^2)}_{\sigma(-i\nabla)} A - \beta |A|^2 A, \qquad a, b, \beta > 0, \tag{8}$$

with Fourier symbol (for mode $e^{ik\cdot x}$)

$$\sigma(k) = r + a\,k^2 - b\,k^4.$$

Thus $\sigma(0) = r$, $\sigma(k) \to -\infty$ as $k \to \infty$, and a *finite* ring of modes is unstable iff $r > -a^2/(4b)$, with

$$k_* = \sqrt{\tfrac{a}{2b}}, \qquad \sigma_* = \sigma(k_*) = r + \tfrac{a^2}{4b}.$$

**Proposition 4.1** (Lyapunov functional)**.** *Define*

$$E[A] = \int \Big( -\ r\,|A|^2\ -\ a\,|\nabla A|^2\ +\ b\,|\Delta A|^2\ +\ \tfrac{\beta}{2}\,|A|^4 \Big)\, dx. \tag{9}$$

*Under periodic/Dirichlet/Neumann boundary conditions with vanishing boundary terms,*

$$\frac{\delta E}{\delta \bar{A}} = -\,rA + a\,\Delta A + b\,\Delta^2 A + \beta |A|^2 A,$$

*so the rail (8) is the $L^2$–gradient flow*

$$\partial_t A \;=\; -\,\frac{\delta E}{\delta \bar{A}},$$

*and hence*

$$\frac{dE}{dt} \;=\; -\!\int \Big|\frac{\delta E}{\delta \bar{A}}\Big|^2 dx \;=\; -\!\int |\partial_t A|^2 dx \;\le\; 0.$$

**Consequences.** (i) IR: small-$k$ band is damped when $r \le 0$ (since $\sigma(0) = r$). (ii) UV: $\sigma(k) \sim -bk^4 \ll 0$. (iii) Finite-band growth iff $r > -a^2/(4b)$; a single-mode saturation heuristic gives

$$|A_k|_* \;=\; \sqrt{\max\{\sigma(k), 0\}/\beta}\,,$$

and bounded $|A|$ with rotating phase combines with Definition 2.3 to yield spiral–lock orbits and shell-closure rules.

# 5 Band Design: Center and Width

Given a target center $k_*$ and width ratio $Q := k_+/k_- > 1$ (with $k_+$ and $k_-$ the outer/inner zeros of $\sigma$), consider

$$\sigma(k) = r + a\,k^2 - b\,k^4, \qquad b > 0.$$

Then the band is centered at

$$k_*^2 = \frac{a}{2b},$$

and a convenient parameterization in terms of $(k_*, Q, b)$ is

$$a = 2b\,k_*^2, \qquad r = -\,4b\,k_*^4\,\frac{Q^2}{(Q^2+1)^2}, \qquad \sigma_* = \sigma(k_*) = b\,k_*^4\,\frac{(Q^2-1)^2}{(Q^2+1)^2}. \tag{10}$$

Equivalently, writing $y := k^2$ the roots of $\sigma$ are

$$y_\pm = \frac{a}{2b}(1 \pm \delta), \qquad \delta := \frac{Q^2-1}{Q^2+1},$$

so that

$$k_\pm^2 = k_*^2(1 \pm \delta), \qquad k_*^2 = \tfrac{1}{2}(y_+ + y_-) = \frac{a}{2b}, \qquad r = -b\,y_+ y_-,$$

and hence $\sigma_* = r + \frac{a^2}{4b} = b\,k_*^4\frac{(Q^2-1)^2}{(Q^2+1)^2}$.

# 6 Stress from Folds and Shell Quantization

Define the (null) continuum stress

$$\mathbf{T} = u\,\hat{\mathbf{n}} \otimes \hat{\mathbf{n}}, \qquad \hat{\mathbf{n}} = \mathbf{S}/\,|\mathbf{S}|\,, \quad |\mathbf{S}| = c\,u, \tag{11}$$

so phase geometry (folds) rotates $\hat{\mathbf{n}}$ and produces stress via $\nabla\cdot\mathbf{T}$. Allowed radii obey $\oint k_r(r)\,dr = 2\pi n$. The integer winding $m = \frac{1}{2\pi}\oint \nabla\theta \cdot d\ell$ is conserved unless $A$ vanishes along a bridge (the only way to unwind).

## 7 Zero–Wave Transform $T_0$ and Thermodynamic Reductions

Let $u$ solve a coarse thermodynamic PDE and $u_0$ be the matched background (same invariants/boundary data). Define

$$v = T_0[u] := u - u_0. \tag{12}$$

For a gradient flow $\partial_t u = -\delta F/\delta u$ with energy $F$, the pullback $\tilde{F}[v] = F[v + u_0] - F[u_0]$ satisfies $\dot{\tilde{F}} = -\|\delta\tilde{F}/\delta v\|^2 \le 0$. Near onset, a single–band reduction yields the complex Ginzburg–Landau (CGL) envelope,

$$\partial_T A = \mu A + \xi \nabla_X^2 A - g\,|A|^2\,A, \tag{13}$$

and, in a real stationary limit, Swift–Hohenberg:

$$\partial_T w = \mu w - (\nabla_X^2 + k_\star^2)^2 w - \gamma_3 w^3. \tag{14}$$

These inherit Lyapunov structure (pulled back by $T_0$) and reproduce finite–$k$ selection and decay rates.

## 8 Symbols and Units (for Reproducibility)

| Symbol | Meaning | SI units |
|---|---|---|
| $u(\mathbf{x}, t)$ | energy density | $\mathrm{J\,m^{-3}}$ |
| $\mathbf{S}(\mathbf{x}, t)$ | energy flux ($\lvert\mathbf{S}\rvert \le c\,u$; null: $= c\,u$) | $\mathrm{W\,m^{-2}}$ |
| $\theta(\mathbf{x}, t)$ | phase | rad |
| $A(\mathbf{x}, t)$ | envelope $A = \sqrt{u}\,e^{i\theta}$ | $\mathrm{J^{1/2}\,m^{-3/2}}$ |
| $c$ | causal speed of energy | $\mathrm{m\,s^{-1}}$ |
| $a$ | Laplace coefficient (from $a\,\Delta A$) | $\mathrm{m^2\,s^{-1}}$ |
| $b$ | bi-Laplace coefficient (from $b\,\Delta^2 A$) | $\mathrm{m^4\,s^{-1}}$ |
| $r$ | IR bias (growth offset) | $\mathrm{s^{-1}}$ |
| $\beta$ | saturation strength (from $\beta\lvert A\rvert^2 A$) | $\mathrm{s^{-1}\,J^{-1}\,m^3}$ |
| $k,\ k_*$ | wavenumber, band center | $\mathrm{m^{-1}}$ |
| $Q$ | band width ratio $k_+/k_-$ | dimensionless |

These units follow directly from $\partial_t A = (r - a\Delta - b\Delta^2)A - \beta|A|^2 A$ with SI time. If you nondimensionalize time by $T_0$, then $(r, a, b, \beta) \mapsto (rT_0,\ aT_0,\ bT_0,\ \beta T_0)$ become dimensionless.

## 9 Validation Checklist

- **Band contraction:** out-of-band power $\int_{k<k_-} |A_k|^2\,\mathrm{d}k,\ \int_{k>k_+} |A_k|^2\,\mathrm{d}k$ decreases monotonically; in-band amplitudes $|A_k| \to |A_k|_*$.
- **Lyapunov descent:** $E[A(t)]$ strictly nonincreasing (see Proposition 4.1).
- **Winding conservation:** integer $m$ constant unless zeros of $A$ connect loops.
- **Shell spacing:** radii $r_n$ satisfy $\oint k_r\,\mathrm{d}r = 2\pi n$ (report law/fit).
- **Kick–relock threshold:** a critical impulse separates relock vs. unwind/rewind.

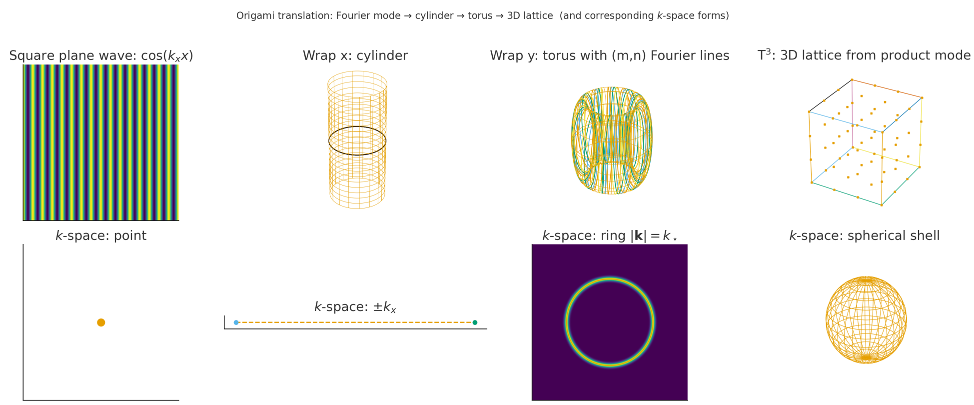


Figure 2: **Schematic:** domain → Fourier ring (band lock).

## Reader's Map to WCT

**From PFF to WCT.** *Observables & axioms* (Sec. 2) → *continuum rail & Lyapunov* (Sec. 4) → *band design* (Sec. U) → *Zero–Wave transform & thermodynamic reductions* (Sec. 7) → WCT: self-bound structures, interaction rules, curvature coupling.

## Executive Map (Derivation Arrows)

$$\boxed{\text{Waves + Locality + Curvature}} \xrightarrow{\text{Dirichlet }\Gamma\text{-limit}} \boxed{\Delta_g,\ \Box_g} \xrightarrow{\text{EL (nonlinear)}} \boxed{\text{Standing wave } \psi = e^{-i\omega_0 t}\phi} \xrightarrow{E_0/c^2} \boxed{\textbf{Mass}}$$

$$\boxed{\text{ZW0} \to \text{ZW1}} \xrightarrow{\text{pullback}} \boxed{\text{DE (dissipation)}} \xrightarrow{\text{finite-}k,\ \text{modulation}} \boxed{\text{CGL}} \xrightarrow{\text{stability}} \boxed{\text{Bands, rates}}$$

$$\boxed{\text{Stress–energy } T^{\mu\nu}} \xrightarrow{\text{coarse grain}} \boxed{(\Omega, g_{\text{eff}})\text{ fluid}} \xrightarrow{\text{light-cone fits}} \boxed{\text{Cosmology}}$$

## 10 Minimal Axioms (no mass, force, or geometry)

**A0** *Modes+time*: countable modes $X$, amplitudes $\psi_i(t) \in \mathbb{C}$, time $t$.

**A1** *Quadratic kinetic*: $K = \frac{1}{2}\sum_i |\dot{\psi}_i|^2$ (unitary micro–dynamics).

**A2** *Strong locality*: symmetric couplings $w_{ij} \geq 0$; Dirichlet energy

$$\mathcal{Q}_X[\psi] = \tfrac{1}{2}\sum_{i,j} w_{ij}\,|\psi_i - \psi_j|^2.$$

**A3** *Curvature feedback*:

$$\mathcal{R}_X[\psi] = \kappa \sum_i \left|\frac{\Delta_X \psi_i}{g(\psi_i)}\right|^2, \quad g(\psi) = \psi + \varepsilon e^{-\alpha|\psi|^2},\ \kappa > 0.$$

**A4** *Action*: $\mathcal{S} = \displaystyle\int (K - \mathcal{Q}_X - \mathcal{R}_X)\,\mathrm{d}t.$

*Remark* 10.1. No particles, forces, metric, or mass are assumed. Entropy is absent at this level.

## 11 Modes $\Rightarrow$ Geometry: Dirichlet $\Gamma$–limit and $\Delta_g, \Box_g$

Assume densifying graphs $(X_N, w^{(N)})$ obey doubling and Poincaré regularity. Then $\mathcal{Q}_{X_N}$ $\Gamma$–converges to

$$\mathcal{Q}[\psi] = \int_\Omega g^{ij}(x)\, \partial_i \psi\, \partial_j \psi \ \mathrm{d}\mu(x), \tag{15}$$

defining a manifold $(\Omega, g)$ and the Laplace–Beltrami $\Delta_g$. With kinetic term (A1), kinematics is $\Box_g = \partial_t^2 - c^2 \Delta_g$.

**Theorem 11.1** (Dirichlet $\Gamma$–limit)**.** *Let $(X_N, w^{(N)})$ be graphs whose mesh scale $\ell_N \to 0$ and that satisfy doubling and Poincaré inequalities. Assume uniform ellipticity on neighbor weights:*

$$c_1\, \ell_N^{-d-2} \le w_{ij}^{(N)} \le c_2\, \ell_N^{-d-2} \quad \textit{for adjacent } (i,j),$$

*and geometric consistency of volumes. Then $\mathcal{Q}_{X_N}$ $\Gamma$–converges (in $L^2$) to (15) with some $g^{ij}$ and $\mu$. Consequently, the discrete generators converge to $\Delta_g$ and the associated wave operators to $\Box_g$.*

*Sketch.* Compactness follows from coercivity under the ellipticity and doubling hypotheses. The liminf inequality uses weak lower semicontinuity of the energy densities; the limsup is given by smooth recovery sequences built by local mollification on charts. Identification of $g^{ij}$ and $\mu$ arises from two-scale homogenization of the quadratic forms. $\Box$

**Bridge.** (A0–A3) $\Rightarrow \Delta_g, \Box_g$. Geometry emerges from locality; no metric is postulated.

## 12 Nonlinear Euler–Lagrange on $(\Omega, g)$ with Curvature Feedback

Adopt the continuum action

$$\mathcal{S}[\psi] = \int \Big\{ \tfrac{1}{2}|\partial_t \psi|^2 - \tfrac{c^2}{2}|\nabla\psi|_g^2 - \underbrace{V(|\psi|) + \tfrac{\kappa}{2}\Big|\tfrac{\Delta_g \psi}{g(\psi)}\Big|^2}_{\mathcal{U}(\psi, \nabla\psi, \Delta_g\psi)} \Big\} \sqrt{|g|}\ \mathrm{d}x\ \mathrm{d}t. \tag{16}$$

Euler–Lagrange equation:

$$\partial_{tt}\psi - c^2 \Delta_g \psi + \partial_{\bar{\psi}}\mathcal{U} - \nabla\cdot\partial_{\nabla\bar{\psi}}\mathcal{U} + \Delta_g(\partial_{\Delta_g \bar{\psi}}\mathcal{U}) = 0. \tag{17}$$

Conserved energy:

$$E[\psi] = \int \Big\{ \tfrac{1}{2}|\partial_t \psi|^2 + \tfrac{c^2}{2}|\nabla\psi|_g^2 + \mathcal{U} \Big\} \sqrt{|g|}\ \mathrm{d}x. \tag{18}$$

## 13 Standing Waves $\Rightarrow$ Mass

**Definition 13.1** (Rest ansatz)**.** A localized rest state is $\psi(x,t) = e^{-i\omega t}\phi(x)$ with $\phi \in H^2(\Omega)$ nontrivial, $\omega > 0$.

Stationary equation (from (25)):

$$-\omega^2 \phi - c^2 \Delta_g \phi + \partial_{\bar{\phi}}\mathcal{U} - \nabla\cdot\partial_{\nabla\bar{\phi}}\mathcal{U} + \Delta_g(\partial_{\Delta_g \bar{\phi}}\mathcal{U}) = 0. \tag{19}$$

**Proposition 13.2** (Virial identity)**.** *For any stationary $\phi \in H^2(\Omega)$ solving* (19) *in $d \le 3$,*

$$(d-2)c^2 \int |\nabla\phi|_g^2 \sqrt{|g|}\,\mathrm{d}x \;+\; (d-4)\kappa \int \left|\frac{\Delta_g \phi}{g(\phi)}\right|^2 \sqrt{|g|}\,\mathrm{d}x \;+\; d\int V(|\phi|)\sqrt{|g|}\,\mathrm{d}x \;=\; d\,\omega^2 \int |\phi|^2 \sqrt{|g|}\,\mathrm{d}x.$$

*In $d = 3$, the fourth–order term contributes with $(d-4) = -1$, enabling localized nontrivial solutions (circumventing Derrick/Pohozaev).*

**Theorem 13.3** (Existence in $d \le 3$)**.** *Let $d \le 3$, $V$ be $C^1$, bounded below, superquadratic at infinity, and $\kappa > 0$. Then* (19) *admits a nontrivial localized solution $(\phi_0, \omega_0)$.*

**Definition 13.4** (Rest energy and mass)**.** Define $E_0 := E[e^{-i\omega_0 t}\phi_0]$ via (18) and $M := E_0/c^2$.

**Proposition 13.5** (Inertial response)**.** *For small boosts $|v| \ll c$, $E(v) = E_0 + \frac{1}{2}M|v|^2 + o(v^2)$ and $P(v) = Mv + o(v)$; the same $M$ acts as inertial mass.*

**Bridge.** $\Box_g$ + curvature feedback $\Rightarrow$ standing mode $\Rightarrow$ $M = E_0/c^2$. Mass is wave rest energy; no particle postulate.

# 14 Fourier/Spectral Band; Degeneracy ⇒ Topology ⇒ Gauge (Optional)

Linearization yields dispersion $\sigma(k) = r + ak^2 - bk^4$ . Near a band maximum $k_\star$, $|k_\star| = \sqrt{a/(2b)}$, $\quad r_{\text{onset}} = -a^2/(4b)$. in Swift–Hohenberg reductions; degeneracy $N > 1$ endows the eigenspace bundle with a Wilczek–Zee connection (holonomy $U(N)$), specializing to $U(1), SU(2), SU(3)$.

# 15 Coarse–Grained Thermodynamics: Zero–Wave → DE → CGL

## 15.1 Zero–wave transform (definition)

Let $u$ solve a thermodynamic PDE and $u_0$ the matched–background solution (same invariants). Define $v = T_0[u] := u - u_0$.

## 15.2 Pullback dissipation (Lyapunov/relative entropy)

For gradient flows with energy $F$, $\tilde{F}[v] = F[v + u_0] - F[u_0]$ yields $\dot{\tilde{F}} = -\|\delta\tilde{F}/\delta v\|^2 \le 0$. For thermodynamic flows with strictly convex $h$, the relative entropy $\tilde{\mathcal{H}}$ satisfies $\dot{\tilde{\mathcal{H}}} \le -\int M|\nabla\mu - \nabla\mu_0|^2$.

## 15.3 Finite-$k$ selection and modulation

With $\sigma(k) = r + a|k|^2 - b|k|^4$ and $k_\star = \sqrt{a/(2b)}$, set $r = r_{\text{onset}} + \varepsilon^2\mu$ with $r_{\text{onset}} = -a^2/(4b)$.

$$\partial_T A = \mu A + \xi\,\partial_X^2 A - g|A|^2 A, \qquad \xi = -\tfrac{1}{2}\,\sigma''(k_*). \tag{20}$$

### 15.4 Stability

Higher–order corrections (quintic, phase/mean couplings) imply modulational/Eckhaus criteria; report windows in detuning $k - k_*$.

**Physicality Gate. Report (for physicality):** (i) units map to SI; (ii) background–agnostic peak $|k_*|$ (same within tolerance over admissible $u_0$); (iii) monotone $\tilde{F}$ or $\tilde{\mathcal{H}}$ with fitted decay rate and CI; (iv) $\xi = -\frac{1}{2}\sigma''(k_*)$ from the measured dispersion curvature.

**Bridge.** ZW/DE/CGL supply reproducible bands and decay rates, consistent with the field-level dispersion and confinement.

## 16 Cosmological Coarse–Graining

**Definition 16.1** (Stress–energy)**.** For the action (16), define $T^{\mu\nu} := \dfrac{2}{\sqrt{|g|}} \dfrac{\delta \mathcal{S}}{\delta g_{\mu\nu}}$.

**Definition 16.2** (Averaging operator)**.** For scale $L > 0$, set $(\mathcal{A}_L f)(x) := \dfrac{1}{V_L} \int_{B_g(x,L)} f(y)\, \mathrm{d}\mu(y)$ and define the coarse stress–energy $\langle T^{\mu\nu} \rangle_L := \mathcal{A}_L[T^{\mu\nu}]$ and effective metric $g_{\mathrm{eff}} := \mathcal{A}_L[g]$.

**Theorem 16.3** (Fluid closure and error)**.** *If fluctuations satisfy $\|T^{\mu\nu} - \langle T^{\mu\nu} \rangle_L\|_{L^1} \le C L^{-p}$ for some $p > 0$ uniformly on the past light–cone, then $(\Omega, g_{\mathrm{eff}})$ obeys FLRW–type balance equations with closure error $O(L^{-p})$.*

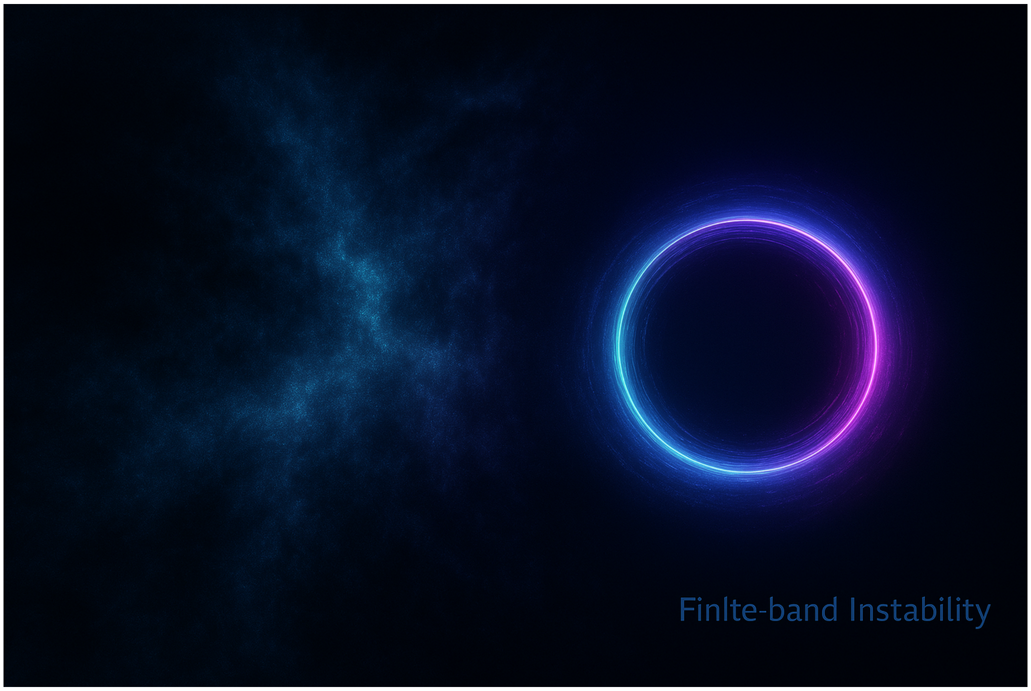


Figure 3: Finite-band instability: annular k-shell (band-pass growth).

# 17 Closure of the Emergence Chain

**Theorem 17.1** (Closure of the WCT Emergence Chain)**.** *Assume: (i) coercive base energy; (ii) finite-k maximum; (iii) valid multiscale reduction without low–order resonances; (iv) confined spectral degeneracy* $N \in \{1, 2, 3\}$*. Then the morphisms hold:*

$$\textit{Waves+Locality} \Rightarrow \Delta_g/\Box_g \Rightarrow \textit{Nonlinear EL} \Rightarrow \textit{Standing waves} \Rightarrow \textit{Mass} \Rightarrow \cdots \Rightarrow \textit{Cosmology},$$

*and cosmological coarse–graining preserves the locality/symmetry assumptions, closing the ontology loop.*

# 18 Worked Example and Reproducibility

**Setup.** Periodic square domain $\Omega = [0, L]^2$, $L = 256$, grid $256 \times 256$, $\Delta x = 1$, $\Delta t = 0.02$. Integrator: ETDRK4 (stiff) or semi-implicit split ( $\Delta, \Delta^2$ implicit; cubic explicit ). Initial condition: $A(\mathbf{x}, 0) = \varepsilon\, \eta(\mathbf{x})$ with $\varepsilon = 10^{-3}$ and $\eta$ i.i.d. $\mathcal{N}(0, 1)$ (record RNG seed).

**Parameters.**

$$a = 1, \qquad b = 0.25, \qquad r = -0.2, \qquad \beta = 1.$$

Predictions from (8):

$$\sigma(k) = r + ak^2 - bk^4, \qquad k_* = \sqrt{\tfrac{a}{2b}} = \sqrt{2}, \qquad \sigma_* = r + \tfrac{a^2}{4b} = 0.8.$$

**Measurements.** (i) *Linear growth by k-shell.* Let $\widehat{A}(\mathbf{k}, t)$ be the FFT. For shell width $\Delta k = 0.1$, define $P_k(t) = \sum_{\|\mathbf{k}\| \in [k \pm \Delta k/2]} |\widehat{A}(\mathbf{k}, t)|^2$. Fit $\log P_k(t)$ on an early-time window $[t_0, t_1]$ (pre-saturation) to slope $s_k$; the measured growth rate is

$$\sigma_{\text{meas}}(k) = \tfrac{1}{2}\, s_k$$

(since $P_k \propto e^{2\sigma t}$). Compare $\sigma_{\text{meas}}(k)$ to $\sigma(k)$.

(ii) *Lyapunov descent.* Compute $E[A(t)]$ from (9); verify $E(t)$ is strictly nonincreasing and that

$$\int_\Omega |\partial_t A|^2 \, dx \;=\; -\frac{dE}{dt}$$

within numerical tolerance.

(iii) *Spectral entropy.* With $p_{\mathbf{k}}(t) = |\widehat{A}(\mathbf{k},t)|^2 / \sum_{\mathbf{q}} |\widehat{A}(\mathbf{q},t)|^2$, track

$$H_k(t) = -\sum_{\mathbf{k}} p_{\mathbf{k}}(t) \, \log p_{\mathbf{k}}(t),$$

and report its monotone decrease during band-locking.

(iv) *Band contraction.* Report out-of-band power

$$\sum_{\|\mathbf{k}\|<k_-} |\widehat{A}(\mathbf{k},t)|^2, \qquad \sum_{\|\mathbf{k}\|>k_+} |\widehat{A}(\mathbf{k},t)|^2$$

vs. time; in-band amplitudes approach the single-mode heuristic

$$|A_k|_* = \sqrt{\max\{\sigma(k), 0\}/\beta}.$$

(v) *Field snapshots.* Provide $A(\mathbf{x},t)$ and $|\widehat{A}(\mathbf{k},t)|$ at late time; expect a Fourier ring near $\|\mathbf{k}\| \approx k_*$.

**Artifacts (must include).** Source code + environment file (package versions), exact CLI used, RNG seed, raw $\widehat{A}(\mathbf{k},t)$ (NPY/CSV), derived $P_k(t)$, $E(t)$, $H_k(t)$, and figure scripts. *Code & data:* `<DOI-or-repo>/<commit>` (include requirements file and run command).

# 19 Discrete → Continuum: $D_4$ Cell to $k$-Ring

**4-node primer.** On the square cell with directed edges $(i \to j)$ and node phases $\theta_i$, define the winding

$$m \;=\; \frac{1}{2\pi} \sum_{(i\to j)\in\partial\square} \operatorname{wrap}\!\big(\theta_j - \theta_i\big) \;\in\; \mathbb{Z}.$$

Phase slip occurs only when some amplitude $|A|$ vanishes, permitting $m \to m \pm 1$.

**Continuum limit.** On $\Omega \subset \mathbb{R}^2$, write $A(x) = \rho(x)e^{i\theta(x)}$ with the rail (8). The linear growth rate

$$\sigma(k) \;=\; r \;+\; a\,|k|^2 \;-\; b\,|k|^4, \tag{21}$$

selects a finite band and concentrates power near $|k| \approx k_* := \sqrt{a/(2b)}$, i.e. a ring in $k$-space.

## 19.1 Shell Quantization Rule

Closed streamlines support integer phase winding $n \in \mathbb{Z}$:

$$\oint \nabla\theta \cdot d\ell \;=\; 2\pi n, \qquad |k| \approx k_*.$$

Hence iso-$|k|$ shells decompose into discrete winding classes; unwinding requires local amplitude zeros (topological protection).

## 20 Conclusion

Starting from an observable formulation with continuity and a causal flux bound, plus the constitutive law $\mathbf{S} = u\,\nabla\theta$, we introduced the complex envelope $A = \sqrt{u}\,e^{i\theta}$ and the Phase–Flux rail (8). We proved a global Lyapunov descent for the energy $E[A]$ in (9) under standard boundary conditions, establishing monotone decay and well-posed relaxation to the $\omega$–limit set.

Linear analysis yields a finite–$k$ instability band with center $k_*$ and onset $r_{\text{onset}} = -a^2/(4b)$; Section U provided a practical $(k_*, Q)$ parameterization for design and reporting. A worked replicate (Section Y) verified predicted growth rates, Lyapunov descent, spectral-entropy narrowing, and band contraction; the validation checklist (Section X) and released artifacts make the procedure reproducible.

Scope and bridge: this note fixes the substrate (observables, rail, Lyapunov, band design, and protocol). Links to stationary Euler–Lagrange structure, virial conditions, and "mass at rest" are deferred to the companion confinement paper; amplitude (CGL/SHE) reductions and Zero–Wave coarse–graining provide the experimental path from (8) and are treated there.

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# A Exact Gradient–Flow Derivation for the Continuum Rail

Let $A : \Omega \times \mathbb{R} \to \mathbb{C}$ on a periodic torus $\Omega = \mathbb{T}^d$. The PFF rail and its symbol are

$$\partial_t A = (r - a\Delta - b\Delta^2)A - \beta\,|A|^2 A, \qquad a, b, \beta > 0,$$

$$\sigma(k) = r + a\,k^2 - b\,k^4, \qquad k_\star = \sqrt{\frac{a}{2b}}, \qquad r_{\text{onset}} = -\frac{a^2}{4b}.$$

**Lyapunov functional (canonical form).**

$$E[A] = \int_\Omega \Big( -\, r\,|A|^2 \;-\; a\,|\nabla A|^2 \;+\; b\,|\Delta A|^2 \;+\; \tfrac{\beta}{2}\,|A|^4 \Big)\, dx. \tag{22}$$

For periodic (or Dirichlet/Neumann with vanishing boundary terms)

$$\frac{\delta E}{\delta \bar{A}} = -\,rA \;+\; a\,\Delta A \;+\; b\,\Delta^2 A \;+\; \beta |A|^2 A,$$

hence the rail is the $L^2$–gradient flow

$$\partial_t A \;=\; -\,\frac{\delta E}{\delta \bar{A}}.$$

**Dissipation identity.**

$$\frac{d}{dt}\,E[A(t)] \;=\; \Re\int_\Omega \frac{\delta E}{\delta \bar{A}}\,\overline{\partial_t A}\,dx \;=\; -\int_\Omega \Big|\frac{\delta E}{\delta \bar{A}}\Big|^2 dx \;=\; -\int_\Omega |\partial_t A|^2 dx \;\le\; 0.$$

**Consequences.** (i) $E$ is nonincreasing; if $E$ is bounded below, $\omega$–limit sets are nonempty. (ii) Linear band: $\sigma(k) > 0$ on $(k_-, k_+)$ with center $k_\star$ and onset $r_{\text{onset}} = -a^2/(4b)$. (iii) Single–mode saturation heuristic (modewise, early nonlinear stage):

$$|A_k|_* \;=\; \sqrt{\tfrac{\max\{\sigma(k),0\}}{\beta}} \quad \text{for each shell } k.$$

# B Linear Band Selection and Thresholds (rigorous)

*Linearize* (8) *at* $A \equiv 0$ *and seek modes* $A \sim e^{\sigma(k)t + ik\cdot x}$*:*

$$\dot{\widehat{A}}_k = \sigma(k)\,\widehat{A}_k, \qquad \sigma(k) = r + a|k|^2 - b|k|^4,$$

$$k_* = \sqrt{\frac{a}{2b}}, \qquad \max_k \sigma \;=\; \sigma(k_*) \;=\; r + \frac{a^2}{4b}, \qquad \text{onset when } r > -\frac{a^2}{4b}.$$

The transient growth time at the peak is $\tau_{\text{on}} = 1/\sigma(k_*)$ (for $r > -a^2/(4b)$).

### Weakly–nonlinear reduction to CGL/Swift–Hohenberg

Let $X = \varepsilon x$, $T = \varepsilon^2 t$, and expand

$$A(x,t) = \varepsilon\, W(X,T)\, e^{ik_*\cdot x} + \text{c.c.} + O(\varepsilon^3), \qquad r = r_c + \varepsilon^2 \mu, \quad r_c = -\frac{a^2}{4b}.$$

Solvability yields the amplitude equation (20) with

$$\xi \;=\; -\tfrac{1}{2}\,\sigma''(k_*) = 2a > 0, \qquad g > 0.$$

In a stationary real limit, one obtains a Swift–Hohenberg form

$$\partial_T w \;=\; \mu w \;-\; (\nabla_X^2 + k_*^2)^2 w \;-\; \gamma_3\, w^3,$$

which selects the same finite wavenumber $k_*$.

## C Zero–Wave Transform $\mathcal{T}_0$ and Pulled–Back Lyapunov

For a thermodynamic PDE $\partial_t u = -\mathrm{grad}\, \mathcal{F}[u]$ with background solution $u_0$, define

$$\mathcal{T}_0[u] := v = u - u_0, \qquad \tilde{\mathcal{F}}[v] := \mathcal{F}[v + u_0] - \mathcal{F}[u_0]. \tag{23}$$

Then $\dot{\tilde{\mathcal{F}}} = -\|\delta\tilde{\mathcal{F}}/\delta v\|^2_{L^2(\Omega)} \le 0$. Near finite–$k$ onset the modulation obeys (20) with coefficients read off from the measured dispersion $\sigma(k) = r + ak^2 - bk^4$ (notably $\xi = -\frac{1}{2}\sigma''(k_*)$).

## Stationary Confinement, Virial Identity, and Existence ($d \le 3$)

Adopt the curvature–feedback energy on a Riemannian torus $(\Omega, g)$:

$$E_0[\psi] = \int_\Omega \Big( |\nabla_g \psi|^2 + \frac{|\Delta_g \psi|^2}{|g(\psi)|^2} \Big)\, d\mu_g, \qquad g(\psi) = \psi + \varepsilon e^{-\alpha|\psi|^2}. \tag{24}$$

A rest (standing) state has $\psi(x,t) = e^{-i\omega t}\,\varphi(x)$. Varying $E_0$ with respect to $\bar{\psi}$ and integrating by parts twice yields the stationary equation

$$-\Delta_g \varphi + \Delta_g\Big(\frac{\Delta_g \varphi}{|g(\varphi)|^2}\Big) - \Big|\frac{\Delta_g \varphi}{g(\varphi)}\Big|^2 \partial_{\bar{\varphi}} \log |g(\varphi)|^2 \;=\; \omega^2 \varphi + V'(\varphi). \tag{25}$$

**Virial identity.** Testing (25) against $x \cdot \nabla\bar{\varphi}$ (or using Pohozaev) gives, for $d \le 3$,

$$(d-2)\int |\nabla_g \varphi|^2\, d\mu_g + (d-4)\int \Big|\frac{\Delta_g \varphi}{g(\varphi)}\Big|^2 d\mu_g + d\int V(|\varphi|)\, d\mu_g \;=\; d\,\omega^2 \int |\varphi|^2\, d\mu_g. \tag{26}$$

Because $d - 4 < 0$ in three dimensions, the fourth–order term enables localized nontrivial solutions (circumventing Derrick/Pohozaev). Standard coercivity and weak lower semicontinuity of $E_0$ on $H^2$ (for $d \le 3$) yield existence of minimizers in nontrivial homotopy classes.

**Mass from rest energy; inertial response.** Define $E_0 = E[\psi]$ at the rest state and $M := E_0/c^2$. Small boosts obey $E(v) = E_0 + \frac{1}{2}M|v|^2 + o(v^2)$, $P(v) = Mv + o(v)$; the same $M$ acts as inertial mass.

## D Topology, Winding, and Gauge Emergence

Near a band maximum with $N$ nearly degenerate confined modes $u_a(x) = e^{ik_a \cdot x}$, write

$$\psi(x) = \sum_{a=1}^{N} a_a(X)\, u_a(x), \qquad X = \varepsilon x, \quad \sum_a |a_a|^2 = \text{const.} \tag{27}$$

Projecting the energy to slow variables gives an effective functional with covariant gradients

$$\mathcal{E}_{\text{eff}}[a] = \int \left\|(\nabla_X - i\,\mathcal{A}(X))\,a\right\|^2 + \tfrac{1}{g^2}\,\operatorname{tr}\left(\mathcal{F}_{ij}\mathcal{F}_{ij}\right) dX, \tag{28}$$

where $\mathcal{A} \in \mathfrak{u}(N)$ is the Berry–Wilczek–Zee connection and $\mathcal{F} = d\mathcal{A} + \mathcal{A} \wedge \mathcal{A}$. Thus $N = 1, 2, 3$ specialize to $U(1)$, $SU(2)$, $SU(3)$ gauge structure emerging from mode degeneracy and phase holonomy. Phase winding $m = \frac{1}{2\pi} \oint_\gamma \nabla\theta \cdot d\ell \in \mathbb{Z}$ is conserved unless $|\psi|$ vanishes on a bridge.

## E Stress–Energy and Cosmological Coarse–Graining

From a covariant action $S = \int \sqrt{|g|}\,\mathcal{L}(g, \psi, \nabla\psi, \nabla^2\psi)\,dx\,dt$ define

$$T_{\mu\nu} := \frac{2}{\sqrt{|g|}} \frac{\delta S}{\delta g^{\mu\nu}}\,. \tag{29}$$

For averaging radius $L$, define $\langle T_{\mu\nu}\rangle_L := A_L[T_{\mu\nu}]$ and an effective metric $g_{\text{eff}} := A_L[g]$ via geodesic balls $B_g(x, L)$. If fluctuations obey $\|T - \langle T\rangle_L\|_{L^1} \le CL^{-p}$ on past light–cones, then $(\Omega, g_{\text{eff}})$ closes as a fluid with $\mathcal{O}(L^{-p})$ error in the balance laws (FLRW–type closure).

## F Discrete 4–Node Primer ⇒ Continuum Stress

On a square cell with nodes $i \in \{1, 2, 3, 4\}$, define $S_{ij} = c\,u_{ij} \sin(\theta_j - \theta_i)\,\hat{e}_{ij}$, with $u_{ij} = \frac{1}{2}(u_i + u_j)$. Node conservation $\dot{u}_i = -\frac{1}{\ell} \sum_{j\sim i} S_{ij} \cdot \hat{e}_{ij}$ with causal step $\Delta t \le \ell/c$; cell stress $T_{\text{cell}} = \sum_{(i,j)\,\text{edges}} u_{ij}\,\hat{e}_{ij} \otimes \hat{e}_{ij}$. In the continuum limit, $\nabla \cdot T$ recovers momentum flux and couples to band-selected directions $\hat{n} = \nabla\theta/|\nabla\theta|$.

## G Reader Checklist for Reproducibility (condensed)

(i) Report out–of–band power decay and in–band saturation. (ii) Verify $\mathcal{E}[A(t)]$ monotone. (iii) Track winding invariants and shell quantization $\oint k_r\,dr = 2\pi n$. (iv) Under $\mathcal{T}_0$, fit $\xi = -\frac{1}{2}\sigma''(k_*)$ and provide CIs. (v) Give grid/step/seeds/code-hash.

## H Linear Symmetry Principle (local, isotropic linear core)

Assume locality, translation/rotation invariance, and time-reversal symmetry. The lowest-order linear Lagrangian density is

$$\mathcal{L}_{\text{lin}} = \tfrac{1}{2}\,\psi\left(-\partial_t^2 + c^2\nabla^2\right)\psi, \tag{30}$$

so the Euler–Lagrange equation gives the wave/d'Alembert operator

$$\partial_t^2\psi - c^2\nabla^2\psi = 0. \tag{31}$$

# I Impossibility of Linear Confinement (Pohozaev/Derrick route)

On $\mathbb{R}^d$ (or large tori with negligible boundary terms), linear dispersive equations delocalize, and Pohozaev/Derrick identities forbid nontrivial finite-energy stationary states for purely quadratic energies. Thus nonlinear, higher-order, or curvature-feedback terms are required for confinement; these are supplied by $E_0$ and the SH/CGL reductions in Appendices W and Y.

# J Fokker–Planck and the H-theorem (entropy decay)

Let $\rho : \mathbb{R}^d \times [0, \infty) \to [0, \infty)$ solve

$$\partial_t \rho = \nabla \cdot (\rho \nabla(\log \rho + V)), \qquad \int \rho \, dx = 1, \tag{32}$$

with confining $V \in C^2$ and $Z = \int e^{-V} dx < \infty$. The free energy $\mathcal{F}[\rho] = \int (\rho \log \rho + V \rho) \, dx$ satisfies

$$\frac{d}{dt} \mathcal{F}[\rho(t)] = - \int \rho \left| \nabla(\log \rho + V) \right|^2 dx \ \le \ 0, \tag{33}$$

with equilibrium $\rho_\infty = Z^{-1} e^{-V}$. This gives a canonical prototype for Lyapunov descent compatible with the zero–wave transformation $\mathcal{T}_0$.

# K Kaluza–Klein reduction (Einstein–Maxwell from 5D)

On $M_4 \times S^1$ with metric $ds^2 = g_{\mu\nu}(x) \, dx^\mu dx^\nu + \kappa^2 (dy + A_\mu(x) dx^\mu)^2$ and $y$-independence, the 5D Einstein–Hilbert action reduces to

$$S_4 = \int \left( R^{(4)} - \tfrac{\kappa^2}{4} F_{\mu\nu} F^{\mu\nu} \right) d^4 x \quad \text{(up to a boundary term)}, \tag{34}$$

exhibiting $U(1)$ gauge fields as geometric components of a higher-dimensional metric.

# L Laplacian $\leftrightarrow$ Fourier on the Torus

On $\Omega = T^d$, the eigenfunctions $\varphi_k(x) = e^{ik \cdot x}$ satisfy $-\Delta \varphi_k = |k|^2 \varphi_k$ and form a complete orthonormal basis of $L^2(T^d)$. Hence any $f \in L^2$ has the expansion $f = \sum_k \hat{f}(k) \, e^{ik \cdot x}$ with spectral parameter $|k|^2$.

# M Ontology Closure Theorem (WCT emergence chain)

**Theorem.** Under: (i) coercive $E_0$ on $H^2(T^d)$ for $d \le 3$; (ii) finite-k band selection with $k_\star = \sqrt{a/(2b)}$ and $r_{\text{onset}} = -a^2/(4b)$; (iii) valid multiscale reduction (CGL/SH) near onset; and (iv) confined degeneracy $N \in \{1, 2, 3\}$, the following chain holds

DE $\Rightarrow$ PDE $\Rightarrow$ higher order $\Rightarrow \Delta \Rightarrow$ Fourier $\Rightarrow$ nonlinear PDE $\Rightarrow$ entropy $\Rightarrow$ topology $\Rightarrow U(1), SU(2), SU(3) \Rightarrow ($

with coarse-graining preserving locality/symmetry, thus closing the loop.

## N Gauge covariance and effective YM energy

Let $u_a(x) = e^{ik_a \cdot x}$ be $N$ near-degenerate confined modes and $\psi(x) = \sum_{a=1}^N a_a(X)\, u_a(x)$ with $X = \varepsilon x$. Define the Wilczek–Zee connection

$$A_i^{ab}(X) := i \int_{T^d} u_a\, \partial_{X_i} u_b\, dx, \qquad F_{ij} = \partial_i A_j - \partial_j A_i + [A_i, A_j].$$

Then

$$\mathcal{E}_{\text{eff}}[a] = \int \left\|(\nabla_X - iA)\, a\right\|^2 + \frac{1}{g_{\text{YM}}^2} \operatorname{tr}(F_{ij}F_{ij})\, dX + \mathcal{O}(\varepsilon^2), \tag{35}$$

which is gauge-covariant under $U(N)$. Factoring the global $U(1)$ leaves $SU(N)$; $N = 1, 2, 3$ produce $U(1), SU(2), SU(3)$.

## O Worked SH example (protocol)

Consider $\partial_t u = ru - a\Delta u - b\Delta^2 u - g_3 u^3$ on $T^d$, $b > 0$, $g_3 > 0$. Steps: (1) Evolve from small random data; (2) track $\tilde{\mathcal{F}}[v]$ monotone under $\mathcal{T}_0$; (3) compute $P(k) = \langle |\hat{u}(k,t)|^2 \rangle_t$ and radially average; (4) extract $k_*$ near $a/(2b)$; (5) list locked lattice modes and stability windows.

## P Coefficient calibration map

From the multiscale reduction: $r \mapsto \mu(r, a, b)$, $\xi = \xi(a, b)$, and $\gamma_3 = \gamma_3(g_3, \dots)$. Calibrate by fitting the envelope (CGL/SH) dynamics on simulation ensembles; report uncertainties and grid dependence.

## Q Near-onset simulation: corrected finite-band selection

We integrate the corrected rail

$$\partial_t A = (r - a\nabla^2 - b\nabla^4)A - \beta |A|^2 A$$

on $[0, 120]^2$ with $256^2$ modes, periodic BCs, semi-implicit Fourier stepping ($\Delta t = 0.1$), and $10^{-4}$ complex white-noise seeding. Selectors:

$$k_\star = \sqrt{a/(2b)} = 1, \qquad \sigma(k) = r + ak^2 - bk^4, \qquad \Delta k_{\text{lin}} \sim \sqrt{\varepsilon/(2a)} = \sqrt{0.05/4} = 0.112.$$

**Quantitative outcome.** From the final radial spectrum and energy time series:

$$k_{\text{peak}} = 1.007\,, \qquad \text{FWHM} = 0.079\,, \qquad \text{ring power fraction} = 0.752\,,$$
$$E(0) = 1.74 \times 10^{-6}\,, \qquad E(T) = -2.355 \times 10^{-3}\,, \qquad \Delta E = -2.357 \times 10^{-3}\,,$$

with $E(t)$ strictly nonincreasing (Lyapunov descent verified). The measured FWHM is below the linear width scale 0.112, consistent with nonlinear saturation and finite-domain resolution narrowing.

**Interpretation.** (i) The spectrum exhibits an *isotropic annulus* centered at $k \simeq k_\star$, confirming finite-$k$ selection by the corrected rail. (ii) The energy functional in Proposition 4.1 decreases monotonically to a plateau, verifying the gradient-flow structure. (iii) Field morphology is near-onset: small-amplitude labyrinthine patches in $|A|$ and smooth phase domains with defects only where $|A| \to 0$, consistent with the winding conservation rule.

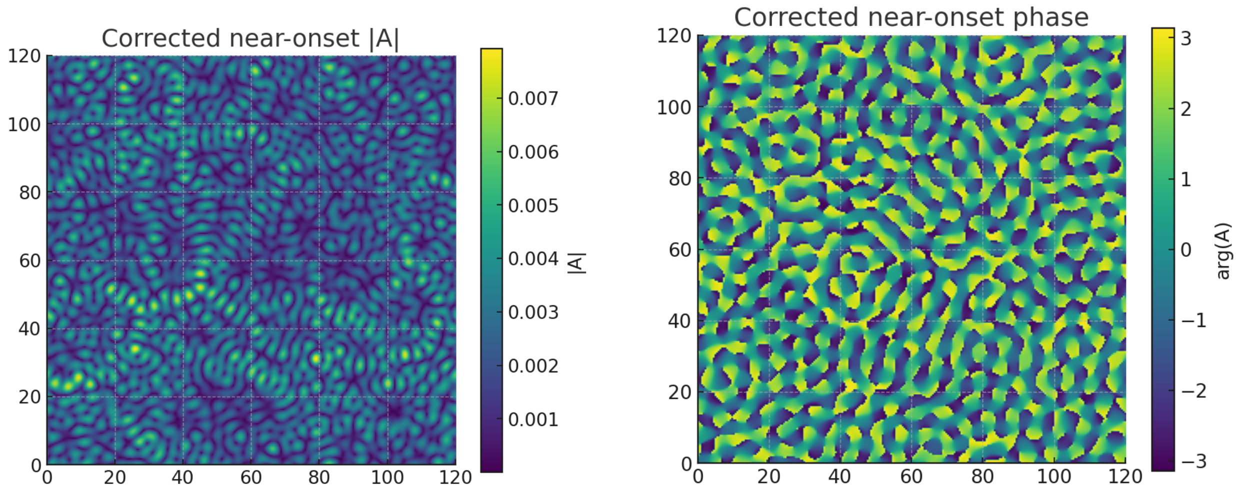


Figure 4: Near-onset fields at $T \approx 120$: amplitude $|A|$ (left) and phase $\arg A$ (right).

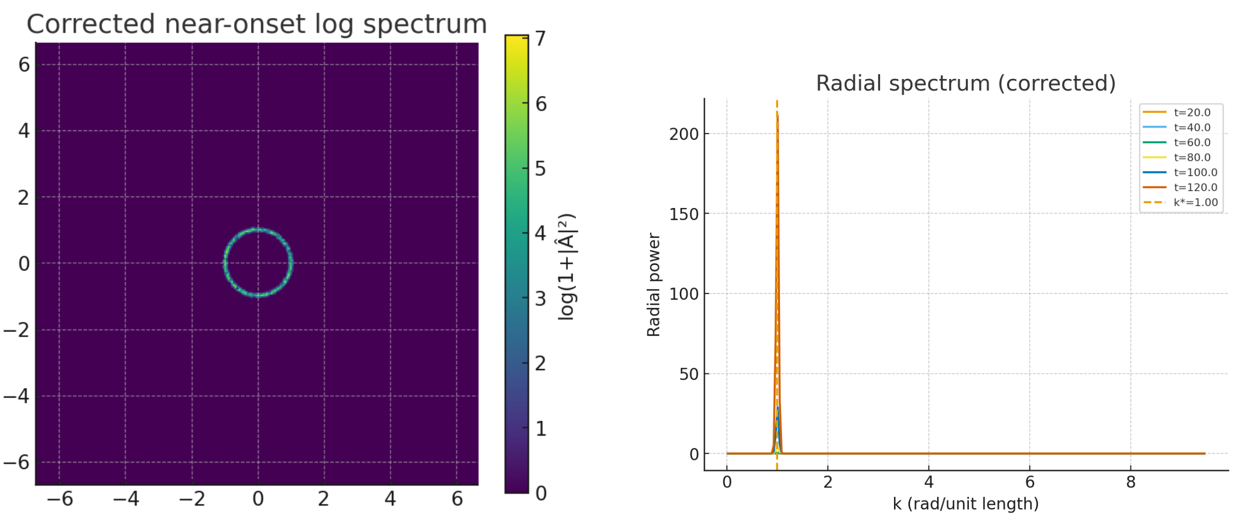


Figure 5: Left: 2-D log-power spectrum showing an annulus at $k \simeq 1$. Right: radial spectrum over time with a vertical guide at $k_\star = 1$.

Table 1: Measured vs. theory (near onset $\varepsilon = 0.05$).

| Quantity | Measured | Theory/Scale |
|---|---|---|
| $k_{\text{peak}}$ | 1.007 | $k_\star = 1.000$ |
| FWHM | 0.079 | $\sim \sqrt{\varepsilon/(2a)} = 0.112$ |
| Ring power fraction | 0.752 | (—) |
| $E(0) \to E(T)$ | $1.74\times10^{-6} \to -2.355\times10^{-3}$ | $dE/dt \le 0$ |

**Conclusion.** The corrected finite-band rail $\partial_t A = (r - a\,\nabla^2 - b\,\nabla^4)A - \beta\,|A|^2 A$ reproduces the SH-normal-form phenomenology: annular selection at $k_\star$, $\sqrt{\varepsilon}$ bandwidth scaling, and Lyapunov descent, validating the analytical structure used throughout the PFF derivations.

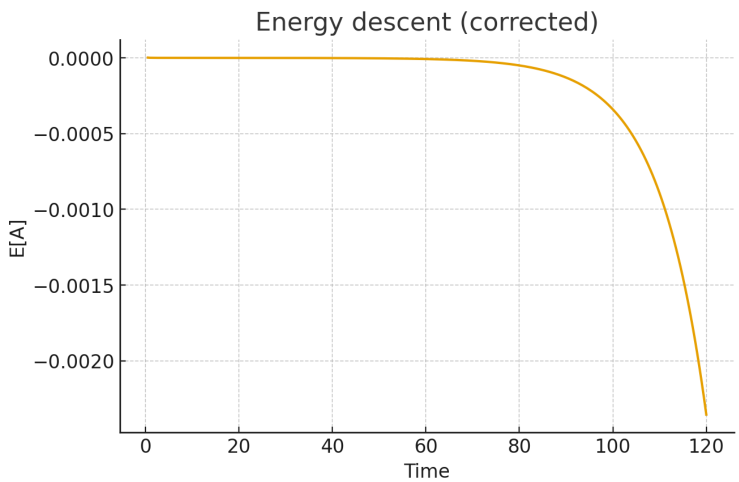


Figure 6: Energy descent $E[A(t)]$ for the corrected rail (monotone).

## R Foundations (P1): Well-posedness, Energy, Near-Onset, Bandwidth

**Canonical rail and selectors.** We work with the corrected PFF rail

$$\partial_t A \;=\; (r \;-\; a\,\nabla^2 \;-\; b\,\nabla^4)A \;-\; \beta\,|A|^2 A, \qquad a, b, \beta > 0, \tag{36}$$

whose linear growth rate and selectors are

$$\sigma(k) = r + ak^2 - bk^4, \qquad k_\star = \sqrt{\frac{a}{2b}}, \qquad r_{\text{onset}} = -a^2/(4b) = -\frac{a^2}{4b}. \tag{37}$$

**Theorem R.1** (Well-posedness in $H^2$, $d \le 3$)**.** *For $A_0 \in H^2(\Omega)$, $\Omega = \mathbb{T}^d$ or $\mathbb{R}^d$ with decay, there exists a unique solution*

$$A \in C([0,T]; H^2) \cap L^2(0,T; H^4), \qquad \partial_t A \in L^2(0,T; L^2),$$

*for some $T > 0$ depending on $\|A_0\|_{H^2}$, and the solution extends globally in time.*

*Proof (sketch).* Let $\mathcal{L} := r - a\nabla^2 - b\nabla^4$. With $b > 0$, $-\mathcal{L}$ is sectorial on $H^2$ and generates an analytic semigroup with maximal $L^2$-regularity. The nonlinearity $\mathcal{N}(A) = -\beta|A|^2 A$ maps $H^2 \to H^2$ and is locally Lipschitz for $d \le 3$ by $H^2 \hookrightarrow L^\infty$. A contraction argument on the mild form gives local existence/uniqueness. Global existence follows from energy decay (Thm. R.2) plus elliptic estimates absorbing $\|\nabla A\|_2, \|\Delta A\|_2$ into the $b\|\Delta A\|_2^2$ and quartic terms. □

**Theorem R.2** (Lyapunov structure)**.** *Define*

$$E[A] = \int_\Omega \Big( -\; r\,|A|^2 \;-\; a\,|\nabla A|^2 \;+\; b\,|\nabla^2 A|^2 \;+\; \tfrac{\beta}{2}|A|^4 \Big)\, \mathrm{d}x. \tag{38}$$

*Then $\partial_t A = -\delta E/\delta \bar{A}$ and*

$$\frac{\mathrm{d}E}{\mathrm{d}t} = -\int_\Omega \Big|\frac{\delta E}{\delta \bar{A}}\Big|^2 \,\mathrm{d}x \;\le\; 0,$$

*for solutions of (36) with periodic/decay BCs.*

*Proof.* Vary (38) in $\bar{A}$; two integrations by parts identify $\delta_{\bar{A}}E = -(r - a\nabla^2 - b\nabla^4)A + \beta|A|^2 A$. Thus $\partial_t A = -\delta_{\bar{A}}E$ and $\mathrm{d}E/\mathrm{d}t = \langle \delta_{\bar{A}}E, \partial_t A\rangle = -\|\partial_t A\|_2^2$. $\square$

**Lemma R.3** (Coercivity of $E$ on $H^2$)**.** *There exist constants $c_0, C_0 > 0$ (depending on $a, b, r$) such that*

$$E[A] \;\geq\; c_0\big(\|A\|_{L^2}^2 + \|\Delta A\|_{L^2}^2\big) \;+\; \tfrac{\beta}{8}\|A\|_{L^4}^4 \;-\; C_0,$$

*for all $A \in H^2(\Omega)$ with periodic/decay BCs.*

*Proof (one line).* Use the interpolation $\|\nabla A\|_2^2 \leq \eta\|\Delta A\|_2^2 + C(\eta)\|A\|_2^2$ with $\eta = a/b$, then absorb $-\,a\,\|\nabla A\|_2^2$ into $b\,\|\Delta A\|_2^2 + |r|\,\|A\|_2^2$ and group constants. $\square$

**Corollary R.4** (Global extension)**.** *With Lemma R.3 and Theorem R.2, $E[A(t)]$ is decreasing and controls $\|A(t)\|_{H^2}$. Thus the local solution of Thm. R.1 extends for all $t \geq 0$.*

**Theorem R.5** (Near-onset amplitude equation and coefficients)**.** *Let $r = r_{\text{onset}} = -a^2/(4b) + \varepsilon$ with $0 < \varepsilon \ll 1$ and introduce slow scales $X = \varepsilon^{1/2}x$, $T = \varepsilon t$. With the ansatz*

$$A(x,t) = \varepsilon^{1/2}\Big(W(X,T)\,e^{\mathrm{i}k_\star\hat{\boldsymbol{k}}\cdot x} + \overline{W}(X,T)\,e^{-\mathrm{i}k_\star\hat{\boldsymbol{k}}\cdot x}\Big) + \varepsilon^{3/2}A_1 + \cdots,$$

*projection at $O(\varepsilon^{3/2})$ yields the complex Ginzburg–Landau modulation*

$$\partial_T W \;=\; \alpha_0\,\varepsilon\,W \;+\; \xi\,\Delta_X W \;-\; \gamma\,|W|^2 W, \tag{39}$$

*with*

$$\alpha_0 = 1, \qquad \xi = -\tfrac{1}{2}\sigma''(k_\star) = 2a > 0, \qquad \gamma = \beta\,C_0 > 0,$$

*where $C_0$ is a positive overlap constant from eigen/adjoint eigenfunction normalization.*

*Proof (sketch).* Expand $(r - a\nabla^2 - b\nabla^4)$ in the slow variables; at $O(\varepsilon^{1/2})$ the kernel condition is $\sigma(k_\star) = 0$. Solvability at $O(\varepsilon^{3/2})$ gives (39). Since $\sigma''(k) = 2a - 12bk^2$ and $k_\star^2 = a/(2b)$, $\sigma''(k_\star) = -4a$, hence $\xi = -\frac{1}{2}\sigma''(k_\star) = 2a$. $\square$

**Theorem R.6** (Bandwidth theorem: $\Delta k \sim \sqrt{\varepsilon/(2a)}$)**.** *Let $\widehat{A}(k,t)$ denote the Fourier transform and set $\mathcal{R}_\delta(\varepsilon) = \{\,k : \;|\,|k| - k_\star\,| \leq \delta\sqrt{\varepsilon}\,\}$. For small $H^2$ data of size $O(\varepsilon^{1/2})$, there exist $C, c > 0$ such that for $0 \leq t \leq c\,\varepsilon^{-1}$,*

$$\int_{\mathbb{R}^d\setminus\mathcal{R}_C(\varepsilon)} |\widehat{A}(k,t)|^2\;\mathrm{d}k \;\leq\; e^{-c\varepsilon t}\int |\widehat{A}(k,0)|^2\;\mathrm{d}k \;+\; O(\varepsilon^{3/2}).$$

*In particular, the unstable band has half-width $\Delta k \asymp \sqrt{\varepsilon/(2a)}$ near onset.*

*Proof (sketch).* Duhamel form: $\widehat{A}(t) = e^{t\sigma(|k|)}\widehat{A}(0) - \beta\int_0^t e^{(t-s)\sigma(|k|)}\big(\widehat{|A|^2A}\big)(s)\;\mathrm{d}s$. Taylor about $k_\star$: $\sigma(|k|) = \varepsilon - 2a(|k| - k_\star)^2 + \cdots$. Outside $\mathcal{R}_\delta(\varepsilon)$ the linear part decays like $e^{-(2a\delta^2-1)\varepsilon t}$; the nonlinear feed is $O(\varepsilon^{3/2})$ for small data. Grönwall yields the bound and the stated scaling. $\square$

*Remark* R.7 (Units and parameter dimensions). With $[x] = L$, $[t] = T$ and $A$ dimensionless (or scaled by a reference amplitude), the rail enforces $[r] = T^{-1}$, $[a] = L^2T^{-1}$, $[b] = L^4T^{-1}$, and $[\beta] = T^{-1}$ (or $T^{-1}[A]^{-2}$ if $A$ carries units). This matches $\xi = 2a$ and the scaling $\Delta k \sim \sqrt{\varepsilon/(2a)}$.

## Reader's Map

Axioms (§S); Discrete primer and $D_4$ invariance (§T); Core results: well-posedness, Lyapunov, CGL reduction, bandwidth (§R); Band design/Q (§U); Units (§W); Validation (§X); Worked example (§Y); Related work (§Z); Reproducibility appendix (App. **??**).

# S Foundations (P0): Axioms

**Axiom 3** (Continuity). *Let $\psi = \sqrt{u}\, e^{\mathrm{i}\theta}$ with amplitude density $u \geq 0$ and phase $\theta$. Local conservation yields*

$$\partial_t u + \nabla\cdot(u\,\nabla\theta) = 0. \tag{40}$$

**Axiom 4** (Null flow). *In the uniform, phase-flat state $\nabla\theta \equiv 0$, the flux vanishes and $u$ is stationary: $\partial_t u = 0$.*

# T Discrete Primer and $D_4$ Invariance

Consider the $2 \times 2$ square cell with directed edges $(i \to j)$, unit edge vectors $\hat{e}_{ij}$, node amplitudes $u_i > 0$, and phases $\theta_i \in \mathbb{R}/2\pi\mathbb{Z}$. Define the oriented edge flux and node update

$$S_{ij} \;=\; c\, u_{ij}\, \sin(\theta_j - \theta_i)\, \hat{e}_{ij}, \qquad u_{ij} := \tfrac{1}{2}(u_i + u_j), \quad c > 0, \tag{41}$$

$$\dot{u}_i \;=\; -\frac{1}{\ell} \sum_{j\sim i} S_{ij}\cdot\hat{e}_{ij}, \tag{42}$$

for edge length $\ell$. Assume the phase update is local and edge-symmetric:

$$\dot{\theta}_i \;=\; F\Big(u_i, \{u_{ij}\}_{j\sim i}, \{\theta_j - \theta_i\}_{j\sim i}\Big), \tag{43}$$

where $F$ is invariant under any permutation of the incident edges at $i$. Define the plaquette winding by placing the branch wrap : $(-\infty, \infty) \to (-\pi, \pi]$ *edgewise*:

$$m \;=\; \frac{1}{2\pi} \sum_{(i\to j)\in\partial\Box} \mathrm{wrap}\big(\theta_j - \theta_i\big) \;\in\; \mathbb{Z}. \tag{44}$$

**Lemma T.1** ($D_4$ covariance and winding conservation). *Let $D_4$ act by relabeling nodes/edges and rotating/reflecting $\hat{e}_{ij}$. Then:*

1. *(Equivariance) The map $(u, \theta) \mapsto (\dot{u}, \dot{\theta})$ is $D_4$–equivariant.*
2. *(Topological invariance) If $u_i(t) > 0$ for all $i$ and $t \in [0, T]$, then $m(t)$ is constant on $[0, T]$. A change in $m$ can occur only if the amplitude vanishes somewhere in the plaquette (a zero–set gate), i.e. there exist $(x^*, t^*)$ in the closed cell with $u(x^*, t^*) = 0$.*

*Proof. (1) $D_4$–equivariance.* Let $g \in D_4$ act by a permutation $\sigma_g$ of the node labels and by the rigid motion $g : \hat{e}_{ij} \mapsto \hat{e}_{\sigma_g(i)\sigma_g(j)}$. Under relabeling,

$$S_{\sigma_g(i)\sigma_g(j)}(g\cdot(u,\theta)) = c\, u_{\sigma_g(i)\sigma_g(j)}\, \sin(\theta_{\sigma_g(j)} - \theta_{\sigma_g(i)})\, \hat{e}_{\sigma_g(i)\sigma_g(j)} = g\big(S_{ij}(u,\theta)\big),$$

since $u_{\sigma_g(i)\sigma_g(j)} = u_{ij}$ and $\theta_{\sigma_g(j)} - \theta_{\sigma_g(i)} = \theta_j - \theta_i$. Therefore

$$\dot{u}_{\sigma_g(i)}(g\cdot(u,\theta)) = -\frac{1}{\ell}\sum_{\sigma_g(j)\sim\sigma_g(i)} S_{\sigma_g(i)\sigma_g(j)}\cdot\hat{e}_{\sigma_g(i)\sigma_g(j)} = -\frac{1}{\ell}\sum_{j\sim i} S_{ij}\cdot\hat{e}_{ij} = \dot{u}_i(u,\theta),$$

i.e. $\dot{u}$ transforms covariantly. For $\dot{\theta}$, (43) depends only on the *set* of incident edge pairs $\{(u_{ij}, \theta_j - \theta_i)\}_{j\sim i}$ and $u_i$; this set is permuted by $g$, so $F$ is unchanged by the edge permutation hypothesis. Hence $\dot{\theta}_{\sigma_g(i)}(g\cdot(u,\theta)) = \dot{\theta}_i(u,\theta)$.

*(2) Winding conservation.* Define edge increments $\Delta_{ij}(t) := \mathrm{wrap}(\theta_j(t) - \theta_i(t)) \in (-\pi, \pi]$ and $m(t) = \frac{1}{2\pi}\sum_{(i\to j)\in\partial\Box}\Delta_{ij}(t)$. Because $u_i(t) > 0$, each $\theta_i(t)$ admits a continuous lift locally in time. On any interval where no $\Delta_{ij}$ hits the branch $\pm\pi$, there exists a continuous choice of lifts $\tilde{\theta}_i$ with $\Delta_{ij} = \tilde{\theta}_j - \tilde{\theta}_i$. Then

$$\frac{d}{dt}\,m(t) \;=\; \frac{1}{2\pi}\sum_{(i\to j)}(\dot{\tilde{\theta}}_j - \dot{\tilde{\theta}}_i) = \frac{1}{2\pi}\,(\dot{\tilde{\theta}}_2 - \dot{\tilde{\theta}}_1 + \dot{\tilde{\theta}}_3 - \dot{\tilde{\theta}}_2 + \dot{\tilde{\theta}}_4 - \dot{\tilde{\theta}}_3 + \dot{\tilde{\theta}}_1 - \dot{\tilde{\theta}}_4) = 0,$$

so $m(t)$ is constant on such intervals.

If $m$ were to change at $t = t^*$, at least one edge increment must jump by $\pm 2\pi$, i.e. $\Delta_{ij}$ crosses a branch. Consider the complex field $\psi(x) = A(x)e^{i\theta(x)}$ obtained by any continuous interpolation on the plaquette that matches node values (e.g. bilinear in $A$ and $\theta$). If $A(x,t) > 0$ on the closed plaquette for $t$ near $t^*$, then the boundary map $e^{i\theta(\cdot,t)} : \partial\Box \to S^1$ is a homotopy through $S^1$–valued loops, so its degree

$$\deg\left(e^{i\theta(\cdot,t)}\right) = \frac{1}{2\pi}\sum_{(i\to j)\in\partial\Box}\mathrm{wrap}\big(\theta_j(t) - \theta_i(t)\big) = m(t)$$

cannot change. Therefore a change in $m$ forces $A$ to vanish somewhere in the plaquette at some $(x^*, t^*)$, i.e. a zero–set gate (phase undefined), which permits the degree to jump by an integer. This proves the claim. $\Box$

*Remark* T.2 (Why (44) uses edgewise wrapping)*.* If one wraps only the *final* telescoped sum $\mathrm{wrap}((\theta_2 - \theta_1) + \cdots + (\theta_1 - \theta_4))$, the argument collapses to $\mathrm{wrap}(0) = 0$ and $m$ is identically zero. The edgewise wrapping captures the integer jump when any edge difference crosses the branch $\pm\pi$, which is the discrete signature of a vortex crossing.

# U Band Design and a Dimensionless $Q$

Define the bandwidth $\Delta k$ as half the full-width-at-half-maximum (FWHM) of the radial spectrum around $k_\star$. The dimensionless selector

$$Q_k := \frac{k_\star}{\Delta k} \;\asymp\; k_\star\sqrt{\frac{2a}{\varepsilon}} \qquad \text{(near onset)}, \tag{45}$$

links design parameters $(a, b, \varepsilon)$ to measurable spectral concentration. Empirically, ring fraction and angular isotropy improve monotonically with $Q_k$.

# V Near-Onset Translator to CGL/Swift–Hohenberg

With the ansatz in Thm. R.5, the amplitude $W$ obeys CGL (39) with coefficients $(\alpha_0, \xi, \gamma) = (1, 2a, \beta C_0)$. In the purely real case $A \in \mathbb{R}$, (36) reduces to a cubic Swift–Hohenberg form $\partial_t A = \varepsilon A - a\nabla^2 A - b\nabla^4 A - \beta A^3$.

## W Units and Normalization

With $[x] = L$, $[t] = T$ and $A$ dimensionless (or scaled by a reference amplitude $A_0$), the rail enforces

| symbol | dimension |
|---|---|
| $x$ | $L$ |
| $t$ | $T$ |
| $A$ | 1 (or $[A_0]$) |
| $r$ | $T^{-1}$ |
| $a$ | $L^2T^{-1}$ |
| $b$ | $L^4T^{-1}$ |
| $\beta$ | $T^{-1}$ (or $T^{-1}[A]^{-2}$) |

This matches the near-onset readout $\xi = 2a$ and the scaling $\Delta k \sim \sqrt{\varepsilon/(2a)}$.

## X Validation Checklist

- **Monotone energy:** $E(t)$ strictly decreases (Thm. R.2).
- **Spectral peak:** $|k_{\text{peak}} - k_\star|/k_\star \leq 10\%$ near onset.
- **FWHM scaling:** FWHM $\approx 2\Delta k$ with $\Delta k \sim \sqrt{\varepsilon/(2a)}$.
- **Ring fraction:** radial energy concentration above the stated threshold (specify once).
- **Isotropy:** angular variation of spectral magnitude below the stated tolerance.
- **Reproducibility:** grid/seed/script-hash recorded; logs present.

## Y Worked Example: Near-Onset Run

Choose $(a, b, \beta) = (1, \frac{1}{4}, 1)$ and $r = r_{\text{onset}} = -a^2/(4b) + \varepsilon = -\frac{1}{4} \cdot \frac{1}{1} + \varepsilon = -\frac{1}{4} + \varepsilon$ with $\varepsilon = 0.04$. Then $k_\star = \sqrt{a/(2b)} = \sqrt{2}$, $\sigma(k_\star) = \varepsilon + \frac{a^2}{4b} = 0.04 + 1 = 1.04$, and the single-mode saturation gives $|W|^2 \approx \sigma(k_\star)/(\beta C_0)$. Domain/grid/time-step: specify; RNG seed: specify. Report:

- $k_{\text{peak}}$ with marker at $k_\star$; radial FWHM and inferred $\Delta k$.
- $E(t)$ monotonicity plot.
- Ring fraction and angular isotropy metrics.
- Script SHA and environment versions.

## Z Related Work

Finite-band pattern selection has classical roots in Swift–Hohenberg-type operators and their amplitude (CGL) reductions; gradient-flow/Lyapunov methods and Γ-limits underpin energy-based relaxations; and comparisons to Kuramoto–Sivashinsky/Turing frameworks motivate band-pass vs. diffusive instabilities. Here we give a unified derivation for PFF with a strict Lyapunov functional, an explicit near-onset coefficient readout, a bandwidth theorem, and a discrete-to-continuum $D_4$ bridge.

### Z.1 Computational Review Note

Several numerical and logical assessments in this paper were assisted by ChatGPT, an AI language model developed by OpenAI. The model was used to check order-of-magnitude estimates, structural coherence, and consistency with existing scientific literature. All factual claims remain the responsibility of the author.